

I. INTRODUCTION

- **KeyWord Spotting(KWS) Challenge:** Achieving both high accuracy and energy efficiency in low-power edge environments.
- **Our Approach:** We propose a lightweight KWS architecture leveraging the inherent energy efficiency of Spiking Neural Networks (SNNs), which we evaluate on the widely-used GSC V2 dataset [1].

II. METHOD

Overall Architecture

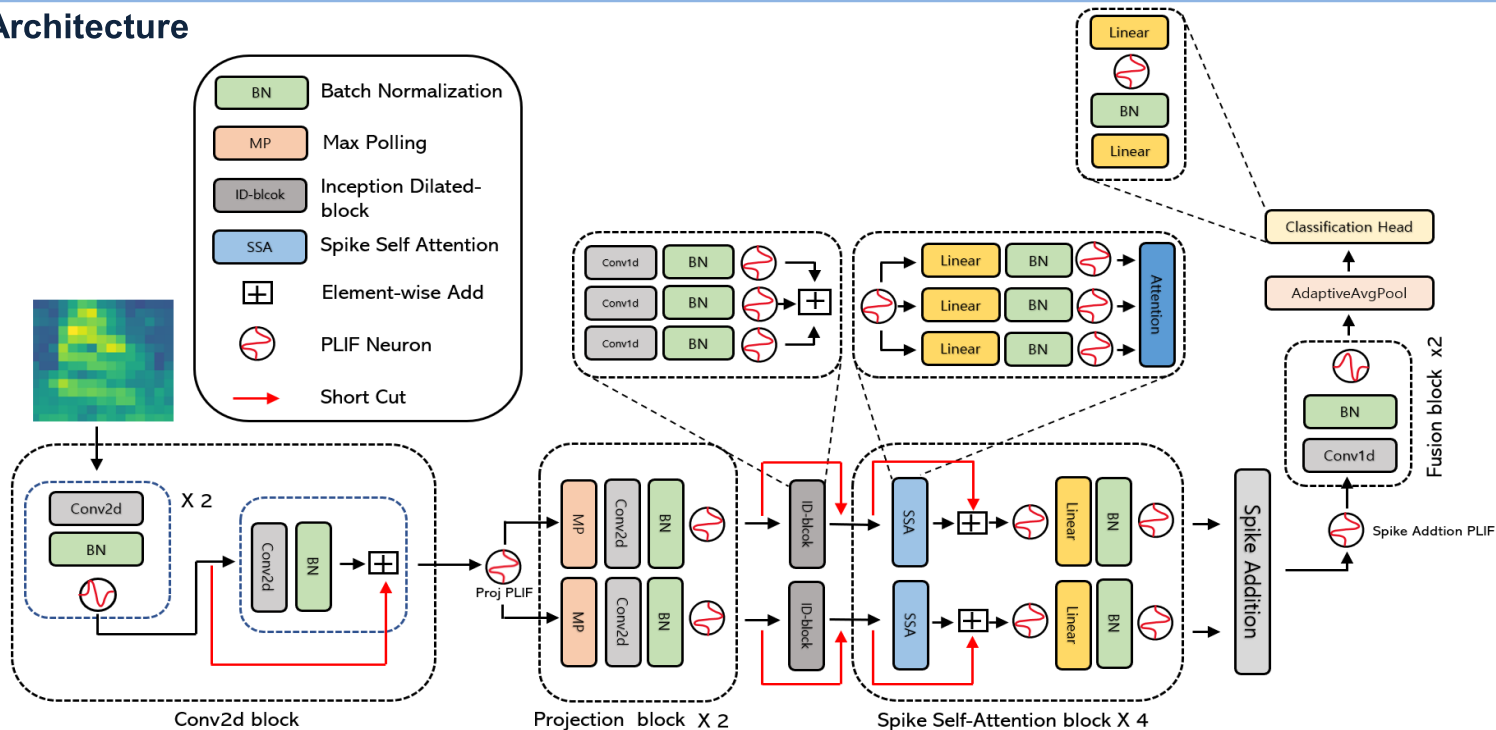


Fig. 1. Overall Architecture

- **Conv2d block:** First captures broad spectral and temporal patterns with a **5x5 kernel**, followed by a refinement step using a **3x3 kernel**.
- **Projection block:** The Projection Block first summarizes the feature map by applying parallel Max-Pooling, then further refines and stabilizes these features using a sequential 3x3 Conv2D and a BatchNorm layer before flattening them into **separate spectral and temporal information**.
- **Inception Dilated(ID) block:** The Inception Block simultaneously captures features from **multiple receptive fields** by employing parallel convolutional paths with **3, 5, and 7 kernels**, and subsequently **combines these multi-scale features**.
- **Spike Self-Attention block:** The Spike Self-Attention block refines the separated spectral and temporal vectors by applying **spike-based self-attention** to each, allowing the model to weigh the **importance of features** and capture their global contextual relationships.

Spike Encoding

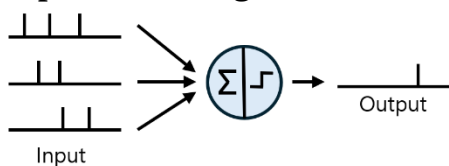


Fig. 2. LIF Neruon [2]

Spike Addition

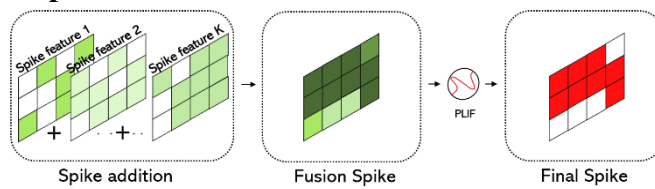


Fig. 3. Spike Addition

Spike Self-Attention [3]

$$Q = LIF(Linear(x))$$

$$K = LIF(Linear(x))$$

$$V = LIF(Linear(x))$$

$$Attention = LIF(QK^T)V$$

III. RESULT

Performance

Table 1. Performance

Model	Model Size (K)	Accuracy (%)
GSC V2 with 12 classes		
SNN-KWS [13]	70.1	94.4
SpikeGRU [19]	111	94.9
SLAYER-RF-CNN [20]	280	91.4
ST-Attention-SNN [21]	2170	95.1
SIDC-KWS (Ours)	401.5	96.8
GSC V2 with 35 classes		
WaveSense [22]	N/A	79.5
LSTMs-SNN [23]	N/A	91.5
SNN-KWS [13]	80.2	92.9
SRNN+ALIF [24]	222.1	92.5
ED-sKWS [14]	306.8	93.1
Speech2Spikes [25]	410	89.5
DCLS-Delays [26]	2500	95.3
SIDC-KWS (Ours)	402.8	94.7

- Achieved 94.7% accuracy with a **model 6x smaller than the SOTA SNN model**, maintaining a performance gap of **less than 1%**.

Spike firing rate

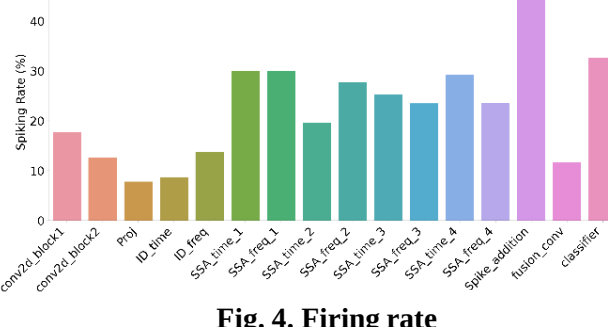


Fig. 4. Firing rate

- **High Overall Efficiency:** Our model achieves a low **average spike firing rate of 21.36%**, indicating significant computational and energy efficiency.
- **Firing Rate Analysis:** **Peak firing activity is localized to Spike Addition operations**, where the aggregation of spikes leads to more frequent threshold crossings.

Ablation study

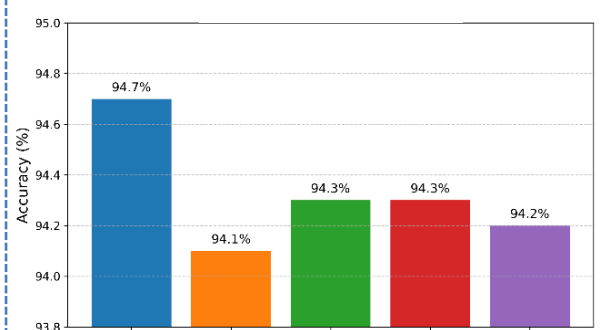


Fig. 5. Ablation Study

1. **ID-Block (Kernel Size):** Verified our multi-scale design by ablating the mixed kernels with uniform **3 and 5** (Fig. 5, orange & green).
2. **Spike Fusion Method:** Evaluated our Spike Addition against alternative fusion methods: **Concatenation and element-wise AND** operations (Fig. 5, red & purple).

IV. CONCLUSION

- **Achieved SOTA performance** for SNNs on GSC V2 (12-class) with 96.8% accuracy (401.5K params).
- Nearly matched the SOTA SNN model on GSC V2 (35-class) (<1% performance gap) with a model ~6x smaller.
- **Demonstrated a 75.59% energy reduction** compared to an equivalent ANN, driven by a low firing rate (21.36%).
- **Future Work:** To validate these efficiency gains on physical neuromorphic hardware.

REFERENCES

1. P. Warden, "Speech commands: A dataset for limited-vocabulary speech recognition," arXiv preprint arXiv:1804.03209, 2018.
2. Wolfgang Maass. Networks of spiking neurons: The third generation of neural network models. Neural Networks, 10(9):1659–1671, 1997.
3. Z. Zhou, Y. Zhu, C. He, Y. Wang, S. Yan, Y. Tian, and L. Yuan, "Spikformer: When spiking neural network meets transformer," in Proc. Int. Conf. Learn. Representations, 2023, pp. 1–17.